1. Re-order level (ROL) $= \text{Maximum consumption} \times \text{Maximum lead-time}$

Or, $= \text{Minimum stock level} + (\text{Daily consumption} \times \text{Normal lead-time})$

Or, $= \text{Safety Stock} + (\text{Daily Consumption} \times \text{Normal Lead time})$

2. Maximum Stock Level $= \text{ROL} + \text{ROQ} - [\text{Min. consumption} \times \text{Min. lead-time})$

3. Minimum Stock Level $= \text{ROL} - [\text{Average consumption} \times \text{Average lead-time})$

Where,

$$\text{Average consumption} = \frac{\text{Max.consumption} + \text{Min. consumption}}{2}$$

$$\text{Average lead-time} = \frac{\text{Max. Lead-time} + \text{Min. Lead-time}}{2}$$

4. $$\text{Average stock level} = \text{Min. Stock Level} + \frac{\text{ROQ/EOQ}}{2} \text{ OR } \frac{\text{Max. Stock Level} + \text{Min. Stock Level}}{2}$$

5. Re-order quantity (ROQ) $= (\text{Max. stock level} - \text{ROL}) + (\text{Min. Consumption} \times \text{Min. Re-order Period})$

6. $$\text{Economic Order Quantity (EOQ)} = \sqrt{\frac{2AO}{C}}$$

7. Total Ordering Cost $= \text{No. of order} \times \text{Ordering cost per order (O) or } \frac{AO}{Q}$

8. Total Carrying Cost $= \text{Average Inventory} \times \text{Carrying Cost per unit (C) or } \frac{QC}{2}$

9. $$\text{No. of Order} = \frac{\text{Annual Requirement (A)}}{\text{EOQ(Q)}}$$

10. $$\text{Total Cost at EOQ (excluding cost of materials)} = \frac{AO}{Q} + \frac{QC}{2} \text{ OR } \sqrt{2AOC}$$

11. $$\text{Total Cost at EOQ (including cost of materials)} = \frac{AO}{Q} + \frac{QC}{2} + PC$$

ILLUSTRATION 18. You are provided the following information of a factory.

Annual demand.................................. 20,000 kg
Fixed charges per order .... Rs. 100
Storage insurance premium10% of inventory value
Rent, rates and taxes....................................50 paisa
Purchase price of material .... Rs. 5 per kg

Required: (a) EOQ (b) Total cost at EOQ

Solution

Given,

Annual Requirement (A) = 20,000 kg

Ordering cost per order (fixed charge) (O) = Rs. 100

Carrying cost per unit (C) $= \text{Rent} + \text{Storage} = 0.50 + 5 \times \frac{10}{100} = \text{Rs. } 1$

(a) We have,

$$\text{EOQ} = \sqrt{\frac{2AO}{C}} = \sqrt{\frac{2 \times 20{,}000 \times 100}{1}} = 2000 \text{ kg}$$

(b) Again,

$$\text{Total Cost at EOQ} = \frac{AO}{Q} + \frac{QC}{2} = \frac{20{,}000 \times 100}{2{,}000} + \frac{2{,}000 \times 1}{2} = 1000 + 1000 = \text{Rs. } 2000$$

= 4,000 units

ILLUSTRATION 22. Material consumption details are:

Normal consumption..........9,000 units per week
Re-order period .... 4-6 weeks
Minimum consumption ..... 6,000 units per week
Maximum consumption... 12,000 units per week

Required: Minimum stock level. **[HSEB 2058]**

Solution

Given,

Normal Consumption per week	= 9000 units	Max. Re-order period	= 6 weeks
Min. Re-order period	= 4 weeks	Minimum Consumption per week	= 6,000 units
Maximum Consumption per week	= 12,000 units		

We have,

Minimum stock level = ROL – (Average consumption × Average re-order period)

$= 72{,}000 - (9{,}000 \times 5) = 27{,}000$ units

Working notes

(i) Re-order level (ROL) = Max. Consumption × Max. Re-order period

$= 12{,}000 \times 6 = 72{,}000$ units

(ii) Average Re-order period $= \dfrac{6+4}{2} = 5$ weeks

ILLUSTRATION 23. A firm provides the following information:

Re-order level ....4,800 units
Daily consumption ....320 to 600 units
Maximum stock level .... 5,000 units
Re-order period .... 4 to 12 days

[HSEB 2059]

Required: Re-order quantity

Solution

We have,

Maximum Stock Level = ROL + ROQ – (Min. Consumption × Min. Lead-time)

Or, 5,000 = 4,800 + ROQ – (320 × 4)

ROQ = 1,480

∴ Re-order quantity = 1,480 units